

Central Coast Regional Water Quality Control Board

July 2, 2019

California Department of Transportation District 5
Shandon Safety Roadside Rest Area
Corby Kilmer, District 5 Coordinator
50 Higuera Street,
San Luis Obispo, CA 93401
Email: corby.kilmer@dot.ca.gov

Sent via Electronic Mail

Dear Ms. Kilmer:

SHANDON SAFETY ROADSIDE REST AREA, HIGHWAY 46, SHANDON, SAN LUIS OBISPO COUNTY – NOTICE OF APPLICABILITY, ENROLLMENT IN ORDER NO. WQ 2014-0153-DWQ GENERAL WASTE DISCHARGE REQUIREMENTS FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS AND TRANSMITTAL OF MONITORING AND REPORTING PROGRAM ORDER NO. R3-2019-0068. TERMINATION OF ENROLLMENT IN ORDER NO. 97-10-DWQ FOR DISCHARGES TO LAND BY SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS

Central Coast Regional Water Quality Control Board (Central Coast Water Board) staff reviewed the available documents associated with upgrades the California Department of Transportation (Caltrans) intends to make to the onsite wastewater treatment and disposal system serving the Shandon Roadside Safety Rest Area (Rest Area) on Highway 46 in northeastern San Luis Obispo County. The Rest Area wastewater treatment system operated by Caltrans is currently enrolled in Water Quality Order No. 97-10-DWQ, General Waste Discharge Requirements for Discharges to Land by Small Domestic Wastewater Treatment Systems (Order 97-10). Order 97-10 was replaced by Order WQ 2014-0153-DWQ, General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems (General Order). The General Order specifies that existing enrollees in Order 97-10 are authorized to continue discharging under Order 97-10 until they are notified by their Regional Water Board that they need to update their coverage. Based on Central Coast Water Board staff's review of your facility information, the updated permit requirements and monitoring and reporting program contained in the General Order are more appropriate for your facility than Order 97-10.

This letter serves as a notice of applicability for your enrollment in the General Order, acknowledges the changes proposed to the wastewater treatment and disposal systems, and provides a new monitoring and reporting program for the Rest Area facility. This letter also serves as a notice of termination of your coverage under Order 97-10.

This letter includes ownership, facility, and facility operation summary information (Attachment 1), contains wastewater treatment system process flow diagrams (Attachment 2), your

monitoring and reporting program requirements (Attachment 3), and a copy of the design plans you submitted (Attachment 4).

Caltrans must comply with the following:

1. **General Order** – Caltrans must comply with all the conditions and requirements of the General Order. As described in the General Order, ongoing operation, maintenance, monitoring, and reporting are required. A copy of the General Order is available electronically at the following link:

https://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2014/wqo2014_0153_dwq.pdf

2. **Monitoring and Reporting Program** – Central Coast Water Board staff evaluated groundwater quality at the project site and determined that additional monitoring for certain constituents was warranted to ensure that groundwater quality objectives are not violated. Caltrans must comply with the requirements of Monitoring and Reporting Program No. R3-2019-0068, attached to this letter. Caltrans is required to submit annual monitoring reports. Your first annual report is due **March 1, 2020** and every year thereafter.

Caltrans is required to submit all requested information electronically in a searchable PDF format by email to CentralCoast@waterboards.ca.gov using the transmittal sheet found at the link below as your cover page:

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/docs/transmittal_sheet.pdf

3. **Fees** – Caltrans paid an annual fee of \$1,004 on January 23, 2018, for coverage in Order No. 97-10. This payment covers Caltrans enrollment in the General Order for the 2018-2019 fiscal year. Caltrans must pay an annual fee to maintain coverage in the General Order. Annual fees are determined by the State Water Resources Control Board's fee program. A copy of the current state fee schedule is available electronically at the following link:

https://www.waterboards.ca.gov/resources/fees/water_quality/docs/fy1819_fee_schedule.pdf

4. **Notification** – The Central Coast Water Board will be notified of your enrollment at a regularly scheduled public meeting on September 19-20, 2019. Details about that meeting are available on our website at:

http://www.waterboards.ca.gov/centralcoast/board_info/agendas/

5. **Technical Report Requirements** – Section E, Provisions (1)(a)(b)(c) of the General Order requires that certain technical reports must be submitted within 90-days of the issuance of this notice of applicability. Failure to submit the required reports in a timely fashion will violate provisions of the General Order.

6. **Future Discharge Modifications** – Pursuant to Water Code section 13260, Caltrans must file a report of waste discharge or secure a waiver from the Executive Officer at least 120 days prior to making any material change or proposed change in the character, location, or volume of your discharge.

Attachment 1 contains Central Coast Water Board staff's understanding of the system at the Rest Area. Please review Attachment 1 to determine if the description is representative of your current system. Pursuant to Water Code section 13260, you must notify Central Coast Water Board staff if this information is out of date or incorrect. You are required to provide staff with your updated information within **30 days** of this letter.

- 7. Responsible Party** – Caltrans is responsible for the management and disposal of the wastewater in compliance with the conditions of the General Order. Any noncompliance with this General Order constitutes a violation of the California Water Code, and subjects Caltrans to enforcement action, and/or termination of enrollment under this General Order.
- 8. Termination of Enrollment** - Caltrans enrollment in Order No. 97-10 is terminated as of the date of this letter.

If you have any questions, please contact **Jon Rokke at (805) 549-3892 or by email at Jon.Rokke@waterboards.ca.gov** or Jennifer Epp at (805) 594-6181 or by email at Jennifer.Epp@waterboards.ca.gov.

Sincerely,

 Digitally signed by Thea Tryon
Date: 2019.07.02 09:12:21
-07'00'


for John M. Robertson
Executive Officer

Attachments:

1. Ownership, Facility, and Facility Operation Summary Information
2. Existing and Modified Wastewater Treatment System Process Flow Diagrams
3. Monitoring and Reporting Program No. R3-2019-0068
4. Design Plans for the Rest Area

cc:

Jon Rokke, Central Coast Water Board, Jon.Rokke@waterboards.ca.gov
Jennifer Epp, Central Coast Water Board, Jennifer.Epp@waterboards.ca.gov
Sheila Soderberg, Central Coast Water Board, Sheila.Soderberg@waterboards.ca.gov
RB3-WDR@waterboards.ca.gov

ECM/CIWQS Place = CW-729031

GT# = WDR100029872

WDR Program

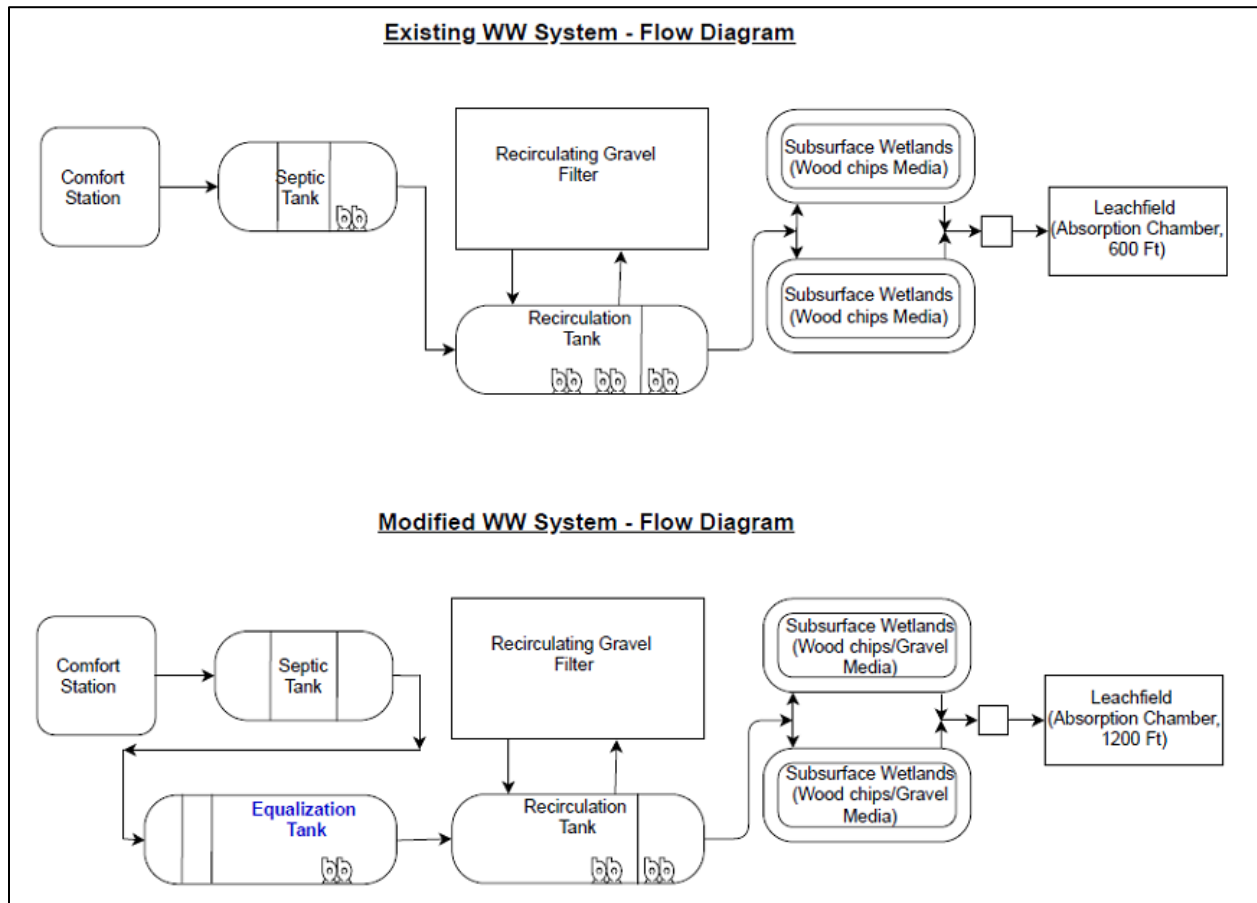
ECM Subject Name = Shandon Safety Roadside Rest Area Notice of Applicability – Small Domestic Wastewater Treatment Systems General Order

R:\RB3\Shared\WDR\WDR Facilities\San Luis Obispo Co\Shandon Safety Roadside Rest Area\2019

Update\Shandon Rest Area Transfer to New GO Letter 7-2-19.docx

ATTACHMENT 1**OWNERSHIP, FACILITY, AND FACILITY OPERATION SUMMARY INFORMATION**

- Shandon Safety Roadside Rest Area (Rest Area) is located on Highway 46 approximately 2.5 miles east of the city of Shandon, California.
- The California Department of Transportation (Caltrans) District 5, Inc. owns and operates the Rest Area.
- Corby Kilmer is the District 5 Rest Area Coordinator.
- Dennis Glickman is the Caltrans District 5 North Region Manager.
- Jerry Marcotte is the Caltrans Senior Sanitary Engineer for this project.
- Bruce O'Hagan is the Caltrans Supervisor for Maintenance at the Rest Area.
- Lee Chaves is the Caltrans District 5 Maintenance Manager.
- Approximately 4,154 gallons per day of wastewater is currently generated at the Rest Area and treated via onsite treatment and disposed systems.
 - o Peak daily flow is 11,545 gallons per day.
 - o Equalized flow to the treatment system is currently 4,585 gallons per day.
- The existing treatment system consists of:
 - o A 25,000-gallon septic tank;
 - o A 3,000 square foot gravel recirculation filter;
 - o A 3,200 square foot subsurface wetland with wood chip media; and
 - o A 600 feet of leach line.
- Changes to the treatment and disposal systems include:
 - o Installation of a 25,000-gallon media/equalization tank split into 3 chambers;
 - The first 2 chambers will contain plastic media; and
 - The third chamber will equalize flows and dose the secondary treatments system at a constant rate.
 - o Changing the media in the subsurface wetland system from wood chips to a combination of wood chips and drainage rock.
 - o Increasing the leachfield area to accommodate 20-year future equalized average daily flows of 5,731 gallons per day.
 - o Changing the pump control system to a Cloud based SCADA system which will provide flow data and pump failure alarms.
- Depth to groundwater is estimated to be 68 feet below ground surface.

ATTACHMENT 2**EXISTING AND MODIFIED WASTEWATER TREATMENT SYSTEM PROCESS FLOW DIAGRAMS**

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
895 Aerovista Place, Suite 101
San Luis Obispo, California 93401**

MONITORING AND REPORTING PROGRAM NO. R3-2019-0068

**for
SHANDON SAFETY ROADSIDE REST AREA
SAN LUIS OBISPO COUNTY**

This Monitoring and Reporting Program (MRP) describes requirements for monitoring a wastewater treatment system for the Shandon Safety Roadside Rest Area. This MRP is issued pursuant to Water Code section 13267. California Department of Transportation District 5 (hereafter, "Discharger") shall not implement any changes to this MRP unless and until a revised MRP is issued by the Central Coast Water Quality Control Board (Central Coast Water Board) Executive Officer.

Water Code section 13267 states, in part:

"In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports."

Water Code section 13268 states, in part:

"(a) Any person failing or refusing to furnish technical or monitoring program reports as required by subdivision (b) of section 13267, or failing or refusing to furnish a statement of compliance as required by subdivision (b) of section 13399.2, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in accordance with subdivision (b).

(b)(1) Civil liability may be administratively imposed by a regional board in accordance with article 2.5 (commencing with section 13323) of chapter 5 for a

violation of subdivision (a) in an amount which shall not exceed one thousand dollars (\$1,000) for each day in which the violation occurs.”

The Discharger owns and operates the wastewater treatment system that is subject to the Notice of Applicability (NOA) of Order WQ 2014-0153-DWQ, General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems (General Order). The burden and cost of preparing the reports is reasonable and consistent with the interest of the state in maintaining water quality. The reports are necessary to ensure that the Discharger complies with the NOA and General Order. Pursuant to Water Code section 13267, the Discharger must implement this MRP and must submit the monitoring reports described herein.

All samples must be representative of the volume and nature of the discharge or matrix of material sampled. The name of the sampler, sample type (grab or composite), time, date, location, bottle type, and any preservative used for each sample shall be recorded on the sample chain of custody form. The chain of custody form must also contain all custody information including date, time, and to whom samples were relinquished. If composite samples are collected, the basis for sampling (time or flow weighted) shall be approved by Central Coast Water Board staff.

Field test instruments (such as those used to test pH, dissolved oxygen, and electrical conductivity) may be used provided that they are used by a State Water Board California Environmental Laboratory Accreditation Program certified laboratory, or:

1. The user is trained in proper use and maintenance of the instruments;
2. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are maintained and available for at least three years.

WATER SUPPLY MONITORING

Representative samples of the water supply must include the following:

Constituent	Units	Sample Type	Sampling Frequency	Reporting Frequency
Total Dissolved Solids	mg/L	Grab	Annually	Annually
Sodium	mg/L	Grab	Annually	Annually
Chloride	mg/L	Grab	Annually	Annually
Total Nitrogen (as N)	mg/L	Grab	Annually	Annually

SEPTIC TANK EFFLUENT MONITORING

Monitoring of septic tank must include the following:

Parameter/ Constituents	Units	Sample Type	Sampling Frequency	Reporting Frequency
Flow Rate	gpd	Metered ^a	Continuous	Annually
BOD ₅	mg/L	Grab	Annually	Annually
Total Suspended Solids	mg/L	Grab	Annually	Annually
Total Dissolved Solids	mg/L	Grab	Annually	Annually
Total Nitrogen (as N)	mg/L	Grab	Annually	Annually

gpd = gallons per day.

BOD₅ = five-day biochemical oxygen demand

a. Flow rate may be metered or estimated based on potable water supply meter readings or other approved method.

Septic tanks shall be inspected and/or pumped at least as frequently as described below. Inspections of sludge and scum depth are not required if the tanks are pumped at least annually.

Parameter	Units	Measurement Type	Inspection/Reporting Frequency
Sludge depth and scum thickness in each compartment of each tank	Feet	Staff Gauge	Annually
Distance between bottom of scum layer and bottom of outlet device	Inches	Staff Gauge	Annually
Distance between top of sludge layer and bottom of outlet device	Inches	Staff Gauge	Annually
Effluent filter condition (if equipped, clean as needed)	NA	NA	Annually

NA denotes not applicable.

Septic tanks shall be pumped when any one of the following conditions exists:

1. The combined thickness of sludge and scum exceeds one-third of the tank depth of the first compartment.
2. The scum layer is within 3 inches of the outlet device.
3. The sludge layer is within 8 inches of the outlet device.

If a septic tank is pumped during the year, the pumping report shall be submitted with the annual report. All pumping reports must be submitted with the next regularly scheduled monitoring report. At a minimum, the record shall include the date, nature of service, service, company name, and service company license number.

EFFLUENT PRIOR TO SOIL DISPERSAL MONITORING

Samples of the effluent (taken prior to disposal area) must include the following:

Constituent	Units	Sample Type	Sampling Frequency	Reporting Frequency
BOD ₅	mg/L	Grab	Annually	Annually
Total Suspended Solids	mg/L	Grab	Annually	Annually
Total Dissolved Solids	mg/L	Grab	Annually	Annually
Total Nitrogen (as N)	mg/L	Grab	Annually	Annually

SOLIDS DISPOSAL MONITORING

The Discharger shall report the handling and disposal of all solids (e.g., screenings, grit, sludge, biosolids, etc.) generated at the wastewater system. Records shall include the name/contact information for the hauling company, the type and amount of waste transported, the date removed from the wastewater system, the disposal facility name and address, and copies of analytical data required by the entity accepting the waste. These records shall be submitted as part of the annual monitoring report.

DISPOSAL AREA

Monitoring must be sufficient to determine if wastewater is evenly applied, the disposal area is not saturated, burrowing animals and/or deep-rooted plants are not present, and odors are not present. Inspection of pump controllers, automatic distribution valves, and any sand or media filter if present is required to maintain optimum treatment in the disposal area. Monitoring must include at a minimum the following:

Parameter	Inspection Frequency	Reporting Frequency
Pump Controllers, Automatic Valves, etc. ^a	Annually	Annually
Nuisance Odor Condition	Annually	Annually
Saturated Soil Conditions ^b	Annually	Annually
Plant Growth ^c	Annually	Annually
Vectors or Animal Burrowing ^d	Annually	Annually

a All pump controllers and automatic distribution valves must be inspected for proper operation as recommended by the manufacturer.

b Inspect the disposal area for saturated conditions. If a mound is used, inspect perimeter base for signs of wastewater seepage or saturated soil conditions.

c Shallow-rooted plants are generally desirable, deep-rooted plants such as trees must be removed as necessary.

d Evidence of animals burrowing must be immediately investigated, and burrowing animal populations controlled as necessary.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, solids, etc.), and reported analytical or visual inspection results are readily discernible. The data must be summarized to clearly illustrate compliance with the General Order and NOA as applicable. The results of any monitoring done more frequently than required at the locations specified in the MRP shall be reported in the next regularly scheduled monitoring report and shall be included in calculations as appropriate.

Annual Report

Annual Reports shall be submitted to the Regional Water Board by **March 1 following the monitoring year**. The Annual Report shall include the following:

1. Tabular and graphical summaries of all monitoring data collected during the year.
2. An evaluation of the performance of the wastewater treatment facility, including discussion of capacity issues, nuisance conditions, system problems, and a forecast of the flows anticipated in the next year. A flow rate evaluation as described in the General Order (Provision E.2.c) shall also be submitted.
3. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into compliance with the NOA and/or General Order.
4. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.
5. The name and contact information for the wastewater operator responsible for operation and maintenance, and system monitoring.

The Discharger is required to submit all requested information electronically in a searchable PDF format by email to CentralCoast@waterboards.ca.gov using a signed transmittal sheet found at the link below as the cover page:

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/docs/transmittal_sheet.pdf

Additionally, electronically submit all reports/documents and laboratory data (using the transmittal sheet as the cover page) to the State Water Resources Control Board's GeoTracker database for the Shandon Roadside Rest Area in San Luis Obispo County site GeoTracker¹ No. WDR100029872 over the internet at:

http://www.waterboards.ca.gov/ust/electronic_submittal/index.shtml

The table below summarizes all the electronic reporting requirements. Staff may request submittal of some documents on paper, particularly drawings or maps that require a

¹ Information for first-time GeoTracker users is available here:
https://www.waterboards.ca.gov/ust/electronic_submittal/docs/beginnerguid2.pdf

large size to be readable, or in other electronic formats where evaluation of data is required.

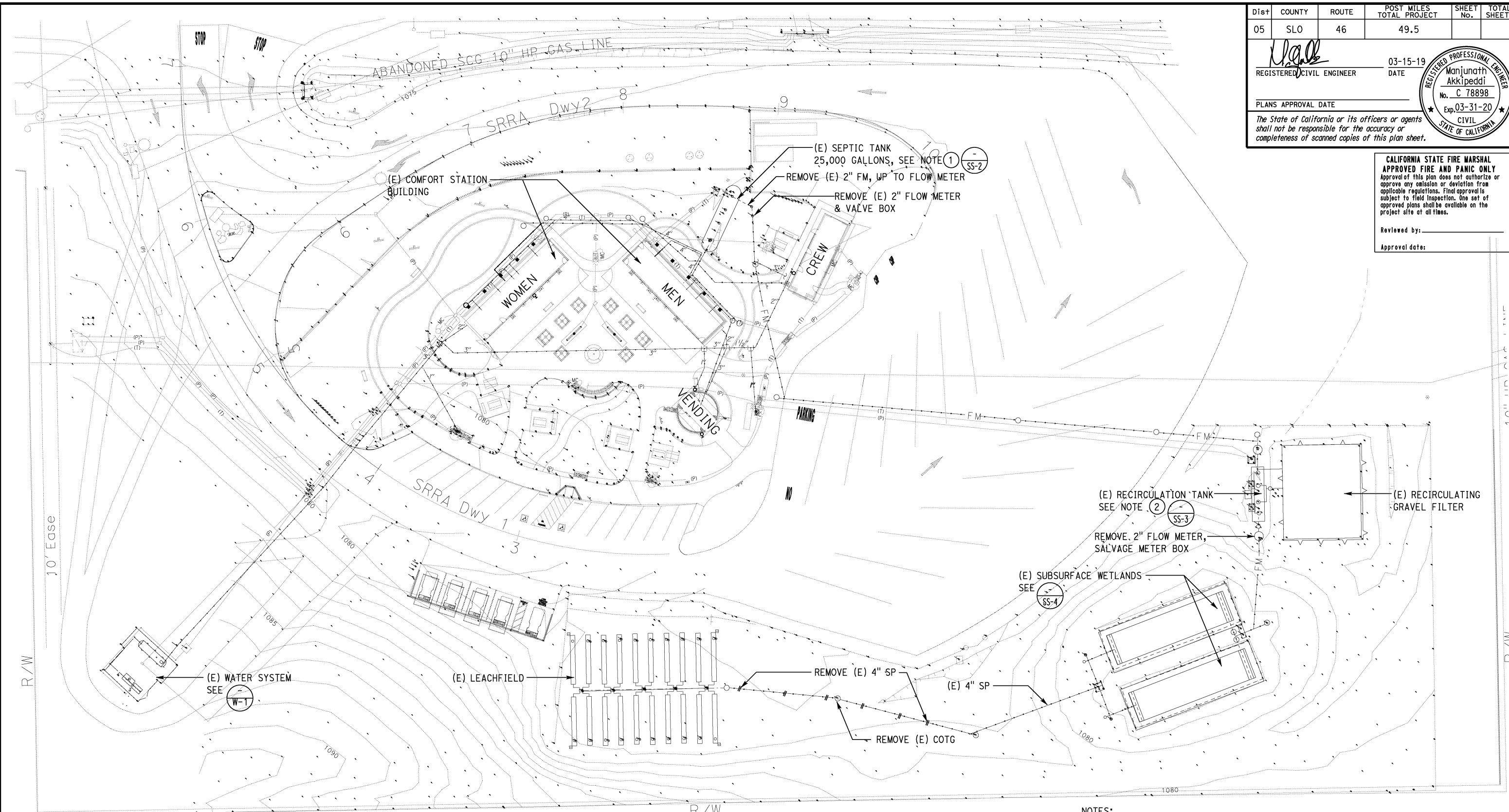
Electronic Submittal	Description of Action	Action	Frequency
Reports and documents	Complete copy of all documents including monitoring reports (in searchable PDF format) and any other associated documents related to the facility.	Upload directly to GeoTracker all monitoring reports (in searchable PDF format) and any other associated documents.	On or before the due dates required by this MRP and for other documents when requested by Central Coast Water Board staff.
Laboratory Data	All analytical data (including geochemical data) in electronic format (EDF). This includes all water, soil, and vapor samples collected when monitoring a discharge.	Direct your State Certified Laboratory staff to upload all laboratory data directly to GeoTracker.	On or before the due date of the required monitoring report.
Location Data (Geo XY)	Survey and mark all permanent sampling locations (i.e., monitoring wells, drinking water wells, and permanent influent/effluent sampling locations). These data points are required prior to laboratory data uploads.	Upload the survey data to the GeoTracker GEO_XY file.	Every time a permanent monitoring point is established.
Geo Map	Site layout, map of facilities, wastewater treatment system, and disposal area(s).	Upload the site layout PDF to the GeoTracker site plan file.	Year one and every five years thereafter and when the facilities are modified.

The Discharger shall implement the above monitoring program as of the date of this MRP.

Ordered by:

  Digitally signed by Thea Tryon
Date: 2019.07.02 09:12:52
-07'00'

for John M. Robertson
Executive Officer



DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		


REGISTERED CIVIL ENGINEER

03-15-19
DATE

PLANS APPROVAL DATE

The State of California or its officers or agents shall not be responsible for the accuracy or completeness of scanned copies of this plan sheet.

CALIFORNIA STATE FIRE MARSHAL
APPROVED FIRE AND PANIC ONLY
Approval of this plan does not authorize or approve any omission or deviation from applicable regulations. Final approval is subject to field inspection. One set of approved plans shall be available on the project site at all times.

Reviewed by: _____
Approval date: _____

EXISTING SITE PLAN

1" = 30'

NOTES:

- ① Pump out sewage/sludge and clean septic tank.
- ② Pump out sewage/sludge and clean recirculation tank.

Prakash Sah DESIGN SUPERVISOR Jerome R. Marcotte DESIGN ENGINEER	DESIGN BY Manjunath Akkipeddi	CHECKED Jerry Marcotte	STATE OF CALIFORNIA DEPARTMENT OF TRANSPORTATION	DIVISION OF ENGINEERING SERVICES ELECTRICAL-MECHANICAL-WATER AND WASTEWATER DESIGN	BRIDGE No. 49R0001	SHANDON SRRA UPGRADE WW SYSTEM, PICNIC SHELTERS EXISTING SITE PLAN	SHEET SS-1
	DETAILS BY Cynthia Pierce	CHECKED Manjunath Akkipeddi			POST MILE 49.5		
QUANTITIES BY Manjunath Akkipeddi			CHECKED Jerry Marcotte	UNIT: 3616 CONTRACT No.: 05-1G3704 PROJECT NUMBER & PHASE: 051500000051	DISREGARD PRINTS BEARING EARLIER REVISION DATES		

TAEMWW Imperial - CCSC Rev. 02/13

ORIGINAL SCALE IN INCHES FOR REDUCED PLANS

0 1 2 3

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08-APR-2019 13:45

Dist	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		


REGISTERED CIVIL ENGINEER

03-15-19
DATE

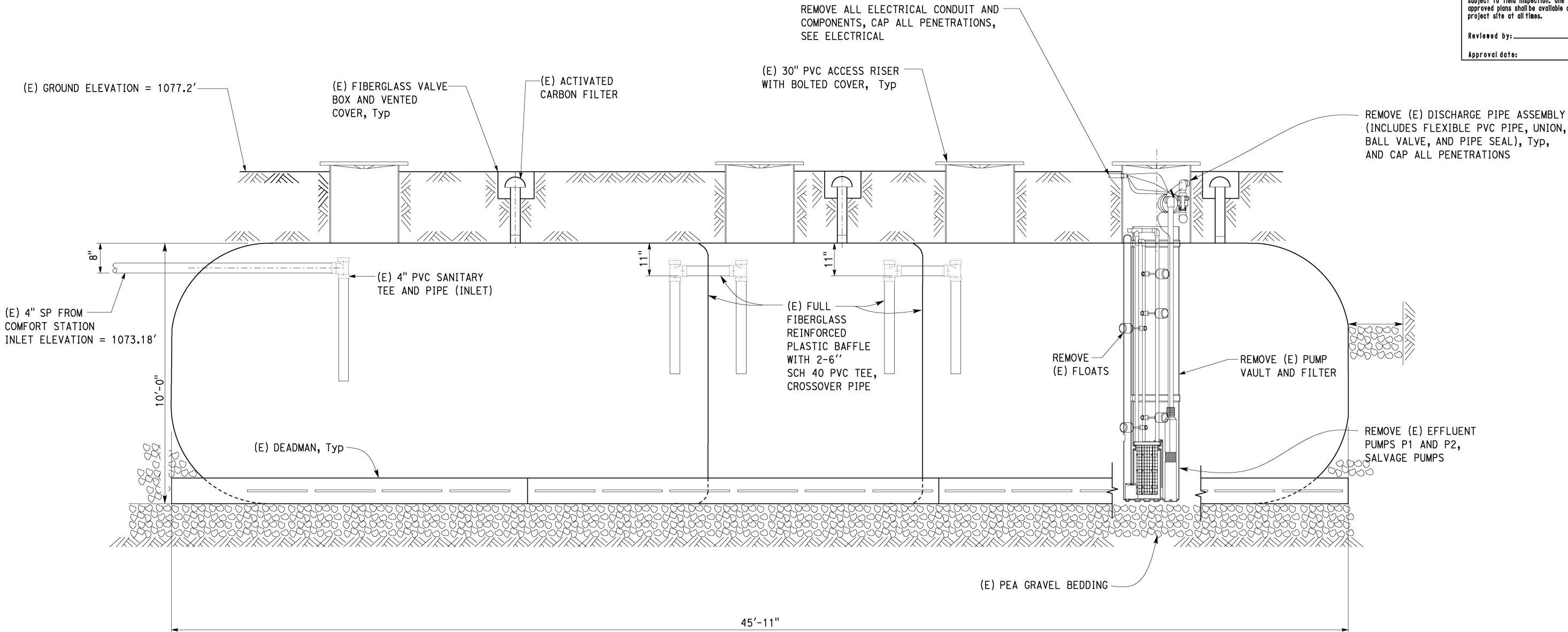
PLANS APPROVAL DATE

The State of California or its officers or agents shall not be responsible for the accuracy or completeness of scanned copies of this plan sheet.

REGISTERED PROFESSIONAL ENGINEER
Manjunath Akkipeddi
No. C 78898
Exp. 03-31-20
CIVIL
STATE OF CALIFORNIA

CALIFORNIA STATE FIRE MARSHAL
APPROVED FIRE AND PANIC ONLY
Approval of this plan does not authorize or approve any omission or deviation from applicable regulations. Final approval is subject to field inspection. One set of approved plans shall be available on the project site at all times.

Reviewed by: _____
Approval date: _____



EXISTING SEPTIC TANK - 25,000 GALLON

NO SCALE

NOTES:

- ① Pump out sewage/sludge and clean septic tank.

DESIGN	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte
DETAILS	BY Cynthia Pierce	CHECKED Manjunath Akkipeddi
QUANTITIES	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte

STATE OF
CALIFORNIA
DEPARTMENT OF TRANSPORTATION

DIVISION OF ENGINEERING SERVICES
ELECTRICAL-MECHANICAL-WATER
AND
WASTEWATER DESIGN

BRIDGE No.
49R0001
POST MILE
49.5

SHANDON SRRA
UPGRADE WW SYSTEM, PICNIC SHELTERS
EXISTING SEPTIC TANK

SHEET
SS-2

Dist	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		


REGISTERED CIVIL ENGINEER

03-15-19
DATE

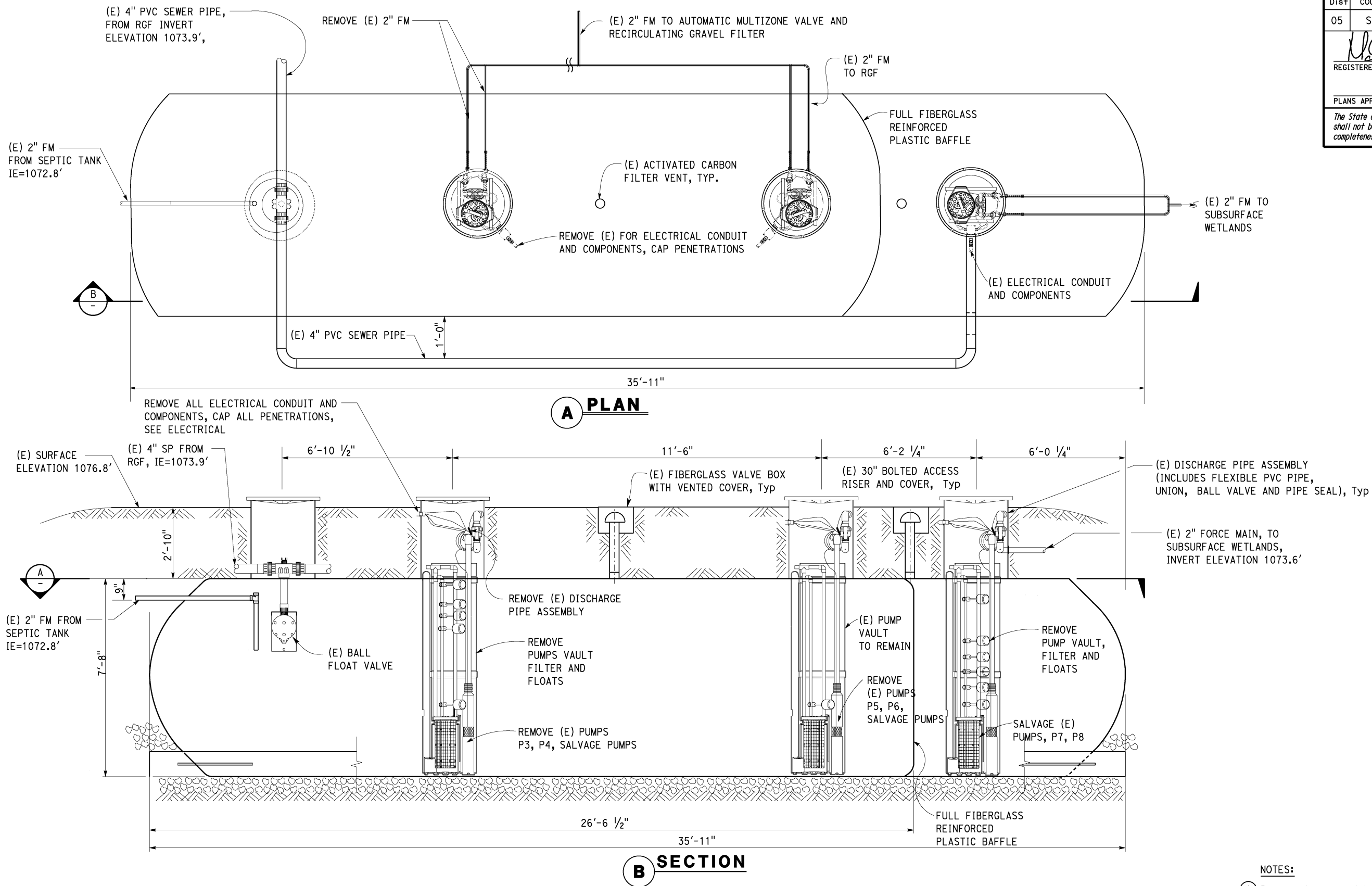
PLANS APPROVAL DATE

Manjunath Akkipeddi
No. C 78898
Exp. 03-31-20
CIVIL
STATE OF CALIFORNIA

The State of California or its officers or agents shall not be responsible for the accuracy or completeness of scanned copies of this plan sheet.

CALIFORNIA STATE FIRE MARSHAL
APPROVED FIRE AND PANIC ONLY
Approval of this plan does not authorize or approve any omission or deviation from applicable regulations. Final approval is subject to field inspection. One set of approved plans shall be available on the project site at all times.

Reviewed by: _____
Approval date: _____



- NOTES:
- 1 Pump out sewage/sludge and clean recirculation tank.

EXISTING RECIRCULATION TANK - 12,000 GALLON

NO SCALE

				DESIGN	BY	Manjunath Akkipeddi	CHECKED	Jerry Marcotte	STATE OF CALIFORNIA DEPARTMENT OF TRANSPORTATION	DIVISION OF ENGINEERING SERVICES ELECTRICAL-MECHANICAL-WATER AND WASTEWATER DESIGN	BRIDGE No.	SHANDON SRRA					SHEET SS-3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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								ORIGINAL SCALE IN INCHES FOR REDUCED PLANS				0				1	2	3	UNIT: 3616 CONTRACT No.: 05-1G3704 PROJECT NUMBER & PHASE: 051500000051	DISREGARD PRINTS BEARING EARLIER REVISION DATES	REVISION DATES (PRELIMINARY STAGE ONLY)					SHEET	OF																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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Dist05COUNTYSLOROUTE46POST MILESTOTAL PROJECT49.5SHEET No.TOTAL SHEETS

03-15-19

REGISTERED CIVIL ENGINEER

03-15-19

DATE

Manjunath Akkipeddi

No. C 78898

Exp. 03-31-20

CIVIL

STATE OF CALIFORNIA

PLANS APPROVAL DATE

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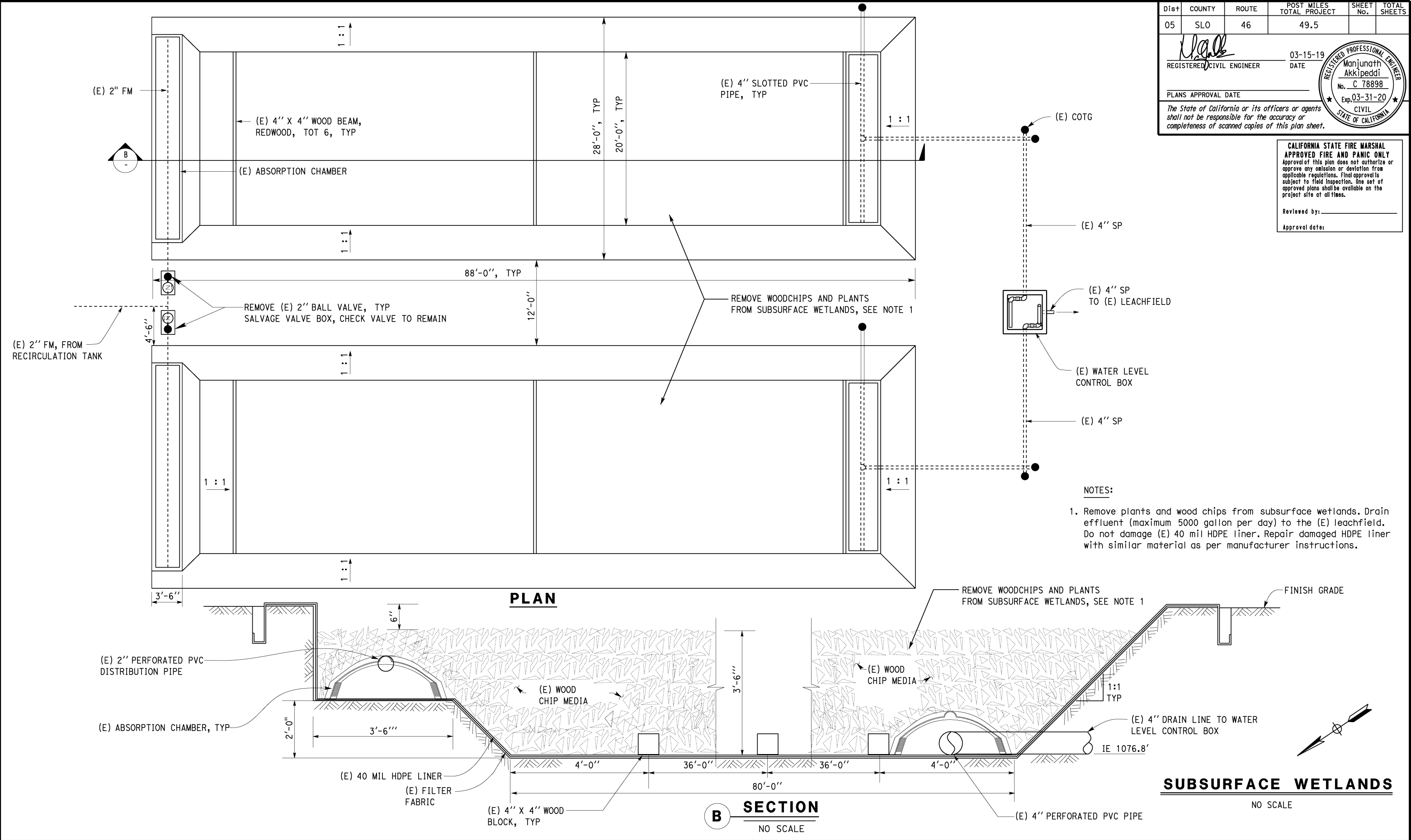
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Reviewed by:

Approval date:



DESIGN BY Manjunath Akkipeddi CHECKED Jerry Marcotte DETAILS BY Cynthia Pierce CHECKED Manjunath Akkipeddi QUANTITIES BY Manjunath Akkipeddi CHECKED Jerry Marcotte			STATE OF CALIFORNIA DEPARTMENT OF TRANSPORTATION		DIVISION OF ENGINEERING SERVICES ELECTRICAL-MECHANICAL-WATER AND WASTEWATER DESIGN		BRIDGE No.		SHANDON SRRA				SHEET		
							49R0001		UPGRADE WW SYSTEM, PICNIC SHELTERS				SS-4		
							POST MILE		EXISTING SUBSURFACE WETLANDS						
						49.5									
ORIGINAL SCALE IN INCHES FOR REDUCED PLANS						0 1 2 3		UNIT: 3616 CONTRACT No.: 05-1G3704 PROJECT NUMBER & PHASE: 051500000051		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)		SHEET OF	
TAEWW Imperial - CCSC Rev. 02/13										03-21-1901-25-1903-15-19					
														P:\dist_05\0515000005_shandon_srra_ww_upgrade\ww\6_FPS&E\ss_4.dgn	

Dist	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		


REGISTERED CIVIL ENGINEER

03-15-19
DATE

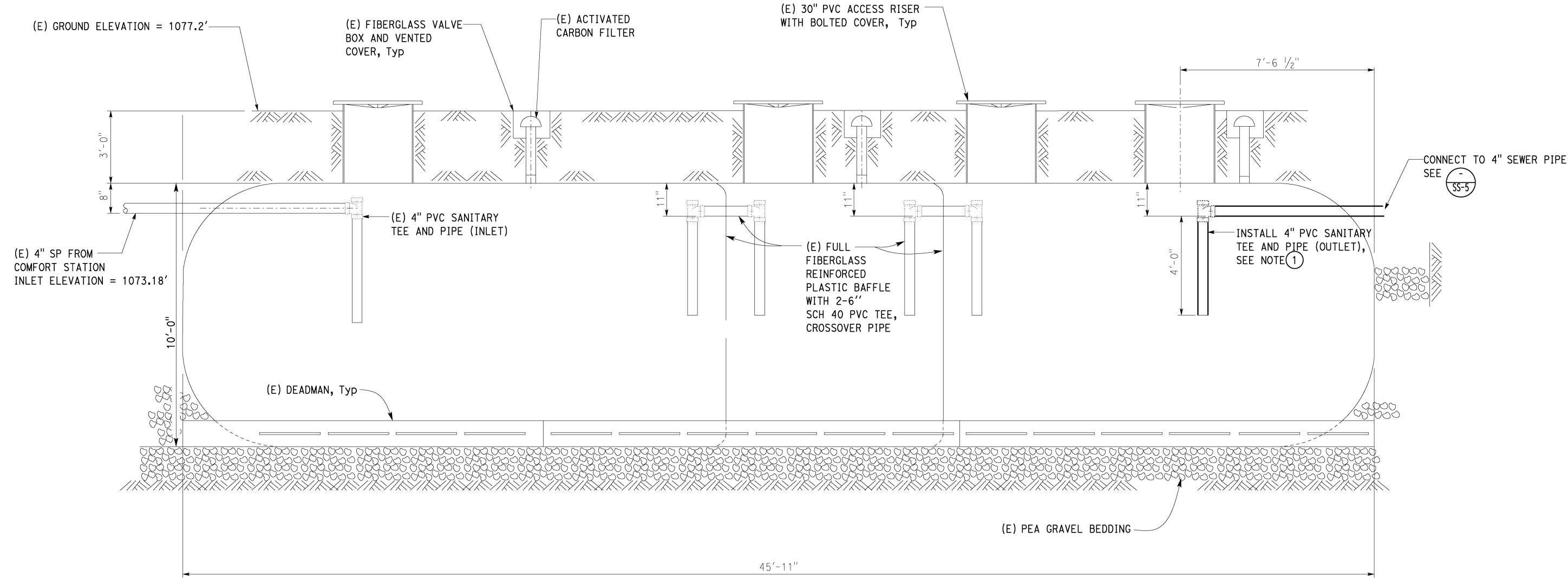
PLANS APPROVAL DATE

Manjunath Akkipeddi
No. C 78898
Exp. 03-31-20
CIVIL
STATE OF CALIFORNIA

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Approval date: _____



MODIFIED SEPTIC TANK - 25,000 GALLON

NO SCALE

NOTES:

- ① DRILL 4" OUTLET HOLE ON THE SEPTIC TANK AND SEAL AS PER TANK MANUFACTURER.

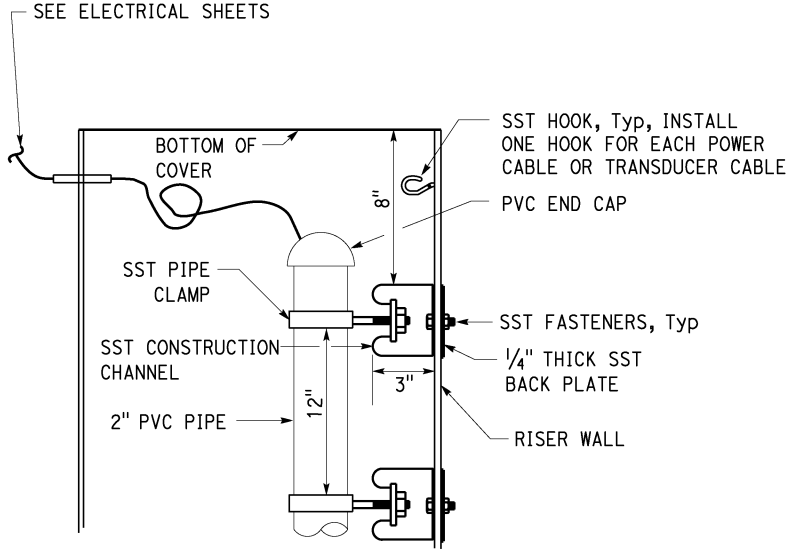
DESIGN BY Manjunath Akkipeddi CHECKED Jerry Marcotte DETAILS BY Cynthia Pierce CHECKED Manjunath Akkipeddi QUANTITIES BY Manjunath Akkipeddi CHECKED Jerry Marcotte				STATE OF CALIFORNIA DEPARTMENT OF TRANSPORTATION		DIVISION OF ENGINEERING SERVICES ELECTRICAL-MECHANICAL-WATER AND WASTEWATER DESIGN		BRIDGE No.		SHANDON SRRA				SHEET			
								49R0001		UPGRADE WW SYSTEM, PICNIC SHELTERS				SS-6			
								POST MILE		MODIFIED SEPTIC TANK							
						49.5											
ORIGINAL SCALE IN INCHES FOR REDUCED PLANS						0 1 2 3		UNIT: 3616 CONTRACT No.: 05-1G3704 PROJECT NUMBER & PHASE: 051500000051		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)				SHEET OF	
TAEWW Imperial - CCSC Rev. 02/13										05-21-1901-25-1903-15-19							
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EQUIPMENT SCHEDULE					
EQUIPMENT	HP	VOLTS	PHASE	PUMPING RATE (GPM)	TOTAL DYNAMIC HEAD (FT)
EFFLUENT PUMPS (EQUALIZATION TANK) <ET-P1> & <ET-P2> USE (E) PUMPS FROM SEPTIC TANK	1/2	230	1	40	40'
PUMP TIMER SETTINGS: PUMP ON = 120 SEC, PUMP OFF = 780 SEC WHEN LIQUID LEVEL IS BETWEEN 27" AND 105"					

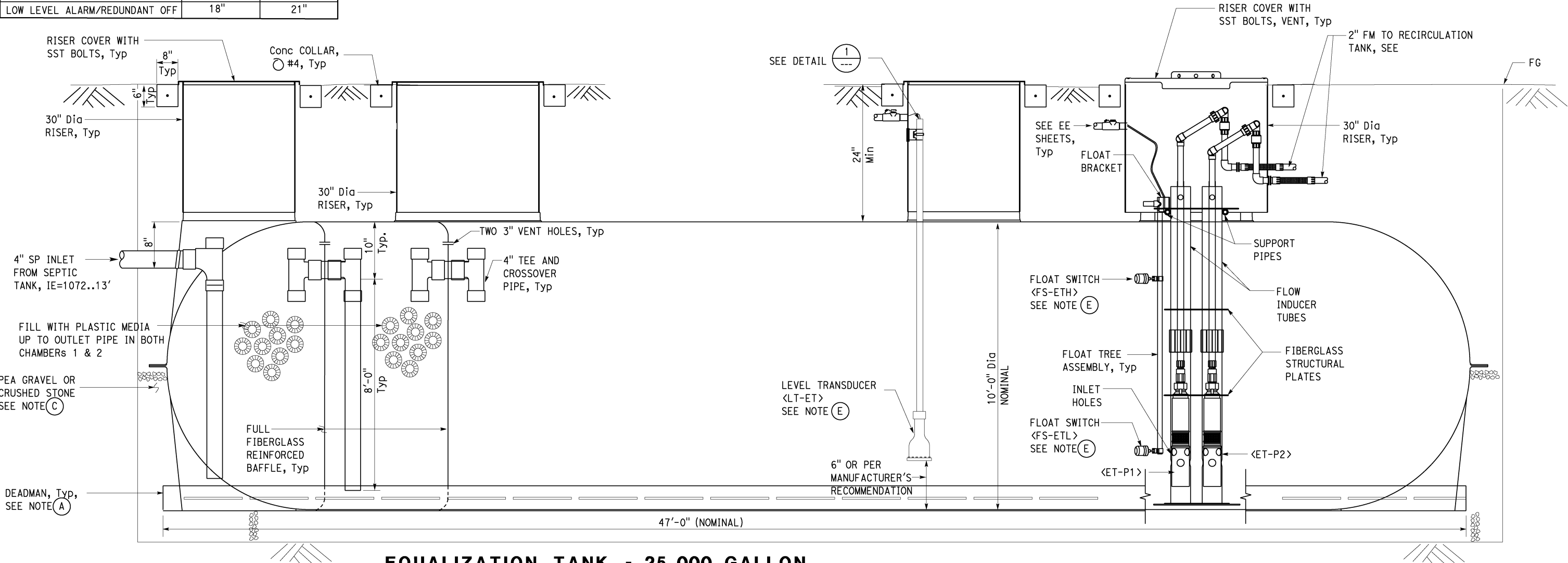
EQUIPMENT OPERATION SCHEDULE		
PUMP OPERATION	FLOAT SWITCH ELEVATION	LEVEL TRANSDUCER ELEVATION
HIGH LEVEL ALARM	-	108"
LAG PUMP ON <FS-ETH> UP	105"	-
LAG PUMP OFF <FS-ETH> DOWN	99"	-
LEAD AND LAG PUMP OFF	-	27"
LOW LEVEL ALARM/REDUNDANT OFF	18"	21"

NOTES:

- (A) Deadman dimensions may vary by manufacturer. Install deadman per tank manufacturer's instructions. Width and height of deadman per manufacturer's recommendation.
- (B) Tank baffles must be installed at locations to divide tank volume into 2500 gallons each for the first and second compartment and 20000 gallons for the third compartment. Lengths may vary by manufacturer.
- (C) Bedding thickness and depth per manufacturer's recommendation.
- (D) Pump piping and electrical lines are skewed in the section view for clarity.
- (E) Xio must provide equipment



1 DETAIL



EQUALIZATION TANK - 25,000 GALLON
NO SCALE

DESIGN	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte
DETAILS	BY Cynthia Pierce	CHECKED Manjunath Akkipeddi
QUANTITIES	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte

STATE OF
CALIFORNIA
DEPARTMENT OF TRANSPORTATION

DIVISION OF ENGINEERING SERVICES
ELECTRICAL-MECHANICAL-WATER
AND
WASTEWATER DESIGN

BRIDGE No.
49R0001
POST MILE
49.5

SHANDON SRRA
UPGRADE WW SYSTEM, PICNIC SHELTERS
EQUALIZATION TANK

SHEET
SS-7

Dist

COUNTY

ROUTE

POST MILES
TOTAL PROJECT

SHEET
No.

TOTAL
SHEETS

05

SLO

46

49.5

Manjunath Akkipeddi

REGISTERED CIVIL ENGINEER

03-15-19

DATE

PLANS APPROVAL DATE

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REGISTERED PROFESSIONAL ENGINEER

Manjunath Akkipeddi

No. C 78898

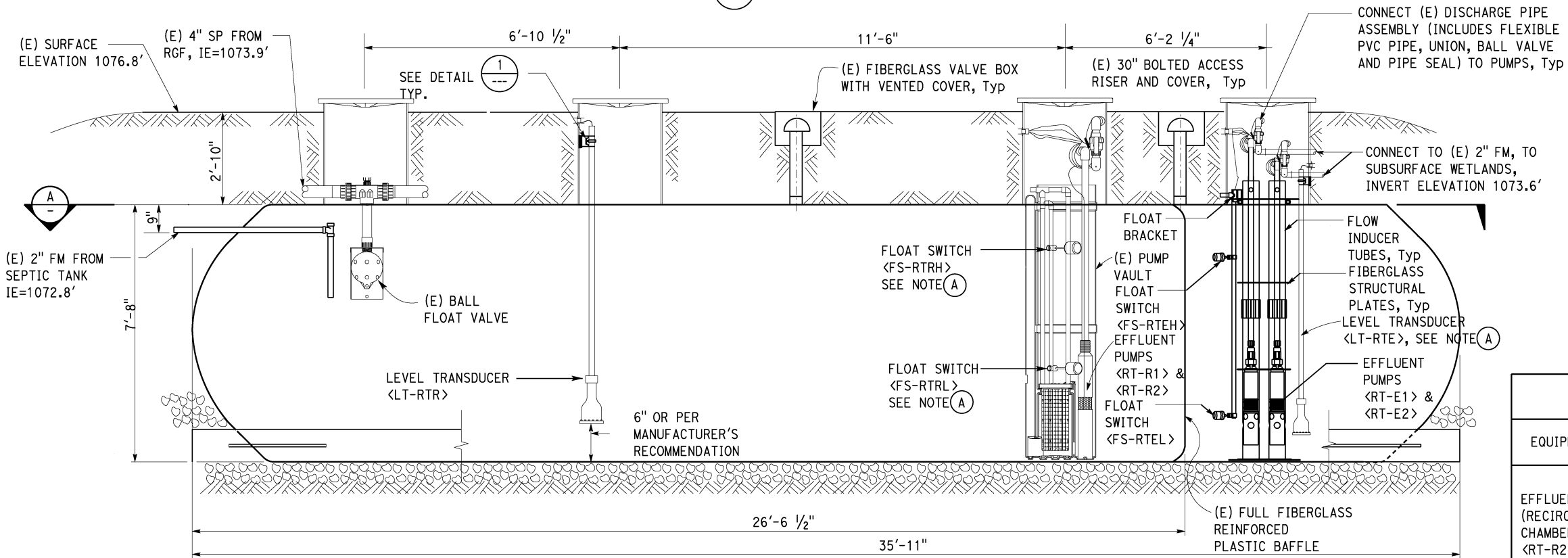
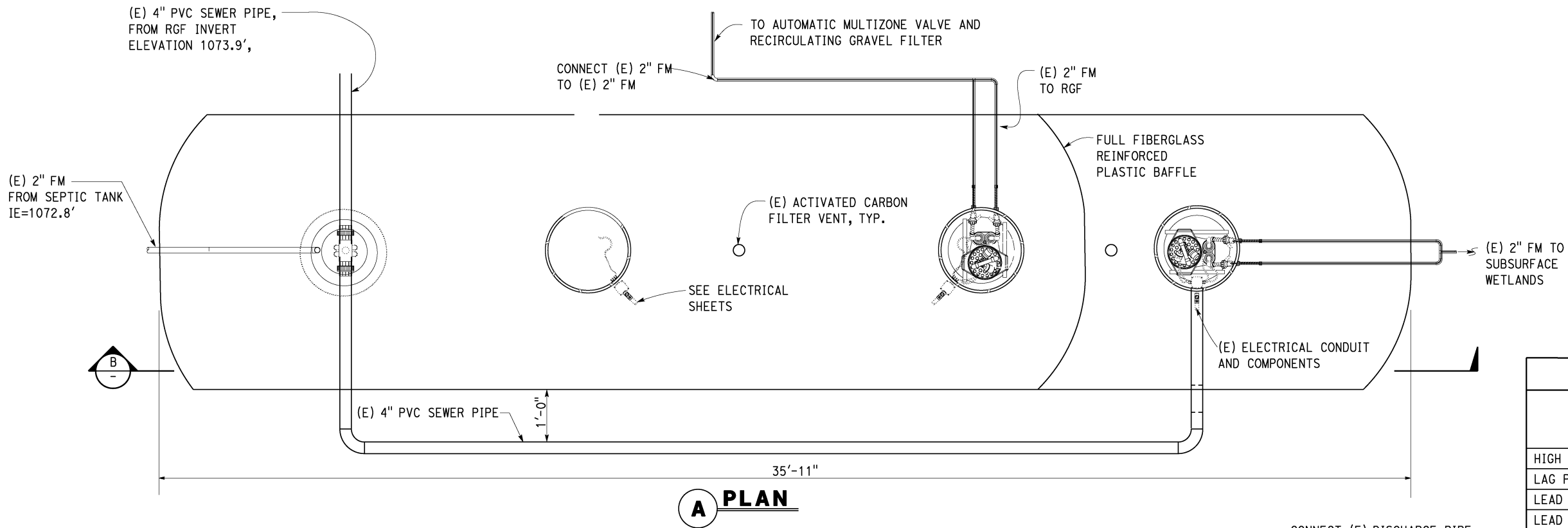
Exp. 03-31-20

CIVIL

STATE OF CALIFORNIA

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NOTES:

(A) Xio must provide equipment

PUMPS RT-E1 & RT-E2 OPERATION SCHEDULE		
PUMP OPERATION	FLOAT SWITCH ELEVATION	LEVEL TRANSDUCER ELEVATION
HIGH LEVEL ALARM	-	87"
LAG PUMP ON	72"	60"
LEAD PUMP ON	-	42
LEAD AND LAG PUMP OFF	-	30
LOW LEVEL ALARM/REDUNDANT OFF	18"	24"

PUMPS RT-R1 & RT-R2 OPERATION SCHEDULE		
PUMP OPERATION	FLOAT SWITCH ELEVATION	LEVEL TRANSDUCER ELEVATION
HIGH LEVEL ALARM	-	87"
LAG PUMP ON <FS-RTRH> UP	80"	-
LAG PUMP OFF <FS-RTRH> DOWN	74"	-
LEAD AND LAG PUMP OFF	-	27"
LOW LEVEL ALARM/REDUNDANT OFF	18"	21"
PUMP TIMER SETTINGS: PUMP ON=300 SEC, PUMP OFF=600 SEC WHEN LIQUID LEVEL IS BETWEEN 27" AND 80"		

EQUIPMENT SCHEDULE					
EQUIPMENT	HP	VOLTS	PHASE	PUMPING RATE (GPM)	TOTAL DYNAMIC HEAD (FT)
EFFLUENT PUMPS (RECIRCULATION CHAMBER) <RT-R1> & <RT-R2>	1	230	1	50	71'
EFFLUENT PUMPS (DISCHARGE CHAMBER) <RT-E1> & <RT-E2> USE (E) PUMPS P7 & P8 FROM DISCHARGE CHAMBER	1/2	230	1	40	40'

MODIFIED RECIRCULATION TANK - 12,000 GALLON

NO SCALE

DESIGN	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte
DETAILS	BY Cynthia Pierce	CHECKED Manjunath Akkipeddi
QUANTITIES	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte

STATE OF
CALIFORNIA
DEPARTMENT OF TRANSPORTATION

DIVISION OF ENGINEERING SERVICES
ELECTRICAL-MECHANICAL-WATER
AND
WASTEWATER DESIGN

BRIDGE No.
49R0001
POST MILE
49.5

SHANDON SRRA
UPGRADE WW SYSTEM, PICNIC SHELTERS
MODIFIED RECIRCULATION TANK

SHEET
SS-8

Dist	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		


REGISTERED CIVIL ENGINEER

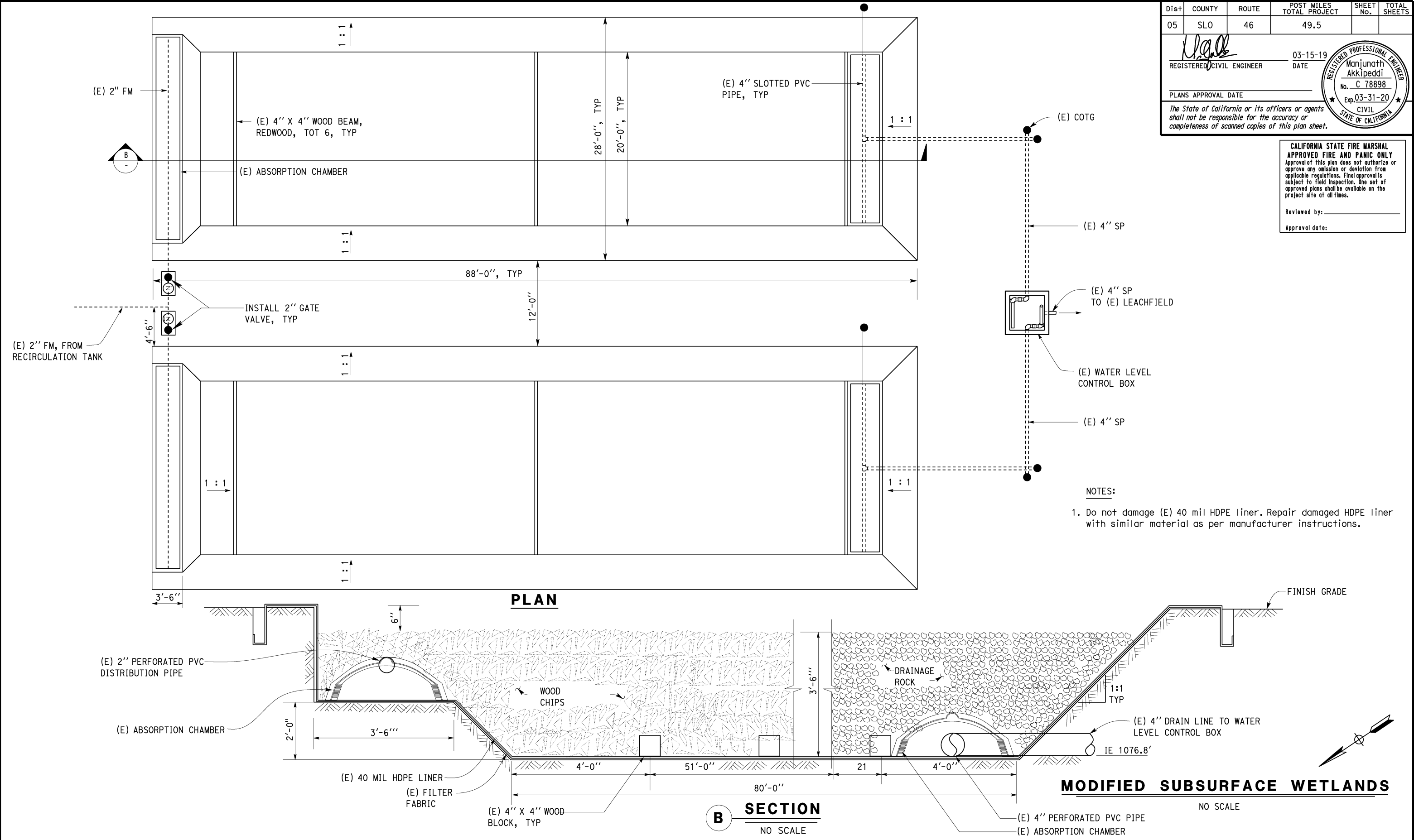
03-15-19
DATE

Manjunath Akkipeddi
No. C 78898
Exp. 03-31-20
CIVIL
STATE OF CALIFORNIA

PLANS APPROVAL DATE _____
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DESIGN BY Manjunath Akkipeddi CHECKED Jerry Marcotte DETAILS BY Cynthia Pierce CHECKED Manjunath Akkipeddi QUANTITIES BY Manjunath Akkipeddi CHECKED Jerry Marcotte				STATE OF CALIFORNIA DEPARTMENT OF TRANSPORTATION		DIVISION OF ENGINEERING SERVICES ELECTRICAL-MECHANICAL-WATER AND WASTEWATER DESIGN		BRIDGE No.		SHANDON SRRA				SHEET			
								49R0001		UPGRADE WW SYSTEM, PICNIC SHELTERS				SS-9			
								POST MILE		MODIFIED SUBSURFACE WETLANDS							
		49.5															
ORIGINAL SCALE IN INCHES FOR REDUCED PLANS						0 1 2 3		UNIT: 3616 CONTRACT No.: 05-1G3704 PROJECT NUMBER & PHASE: 051500000051		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)				SHEET OF	
TAEWW Imperial - CCSC Rev. 02/13										05-21-1901-25-1903-15-19							
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Dist	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		


REGISTERED CIVIL ENGINEER

03-15-19
DATE

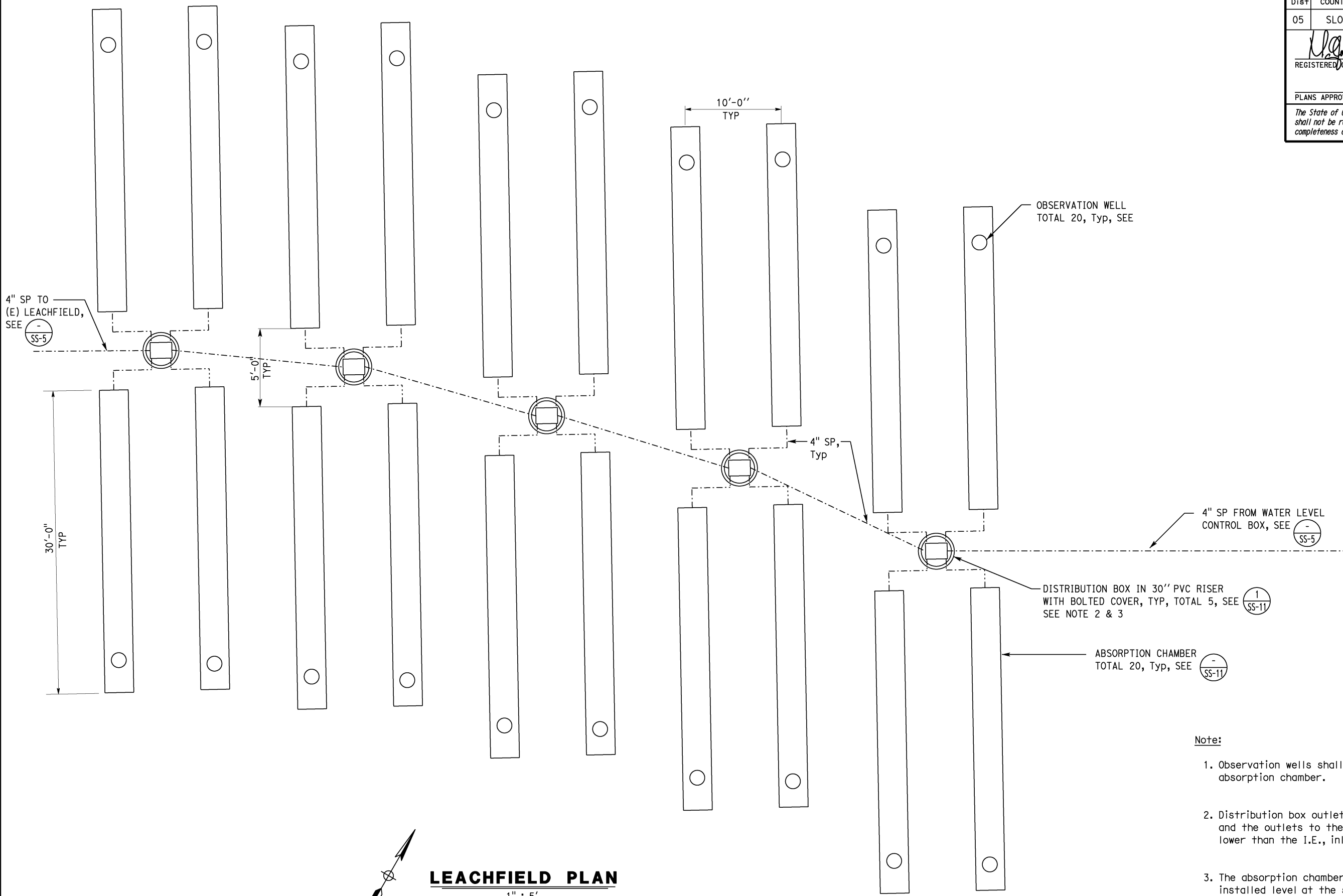
PLANS APPROVAL DATE

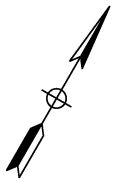
Manjunath Akkipeddi
No. C 78898
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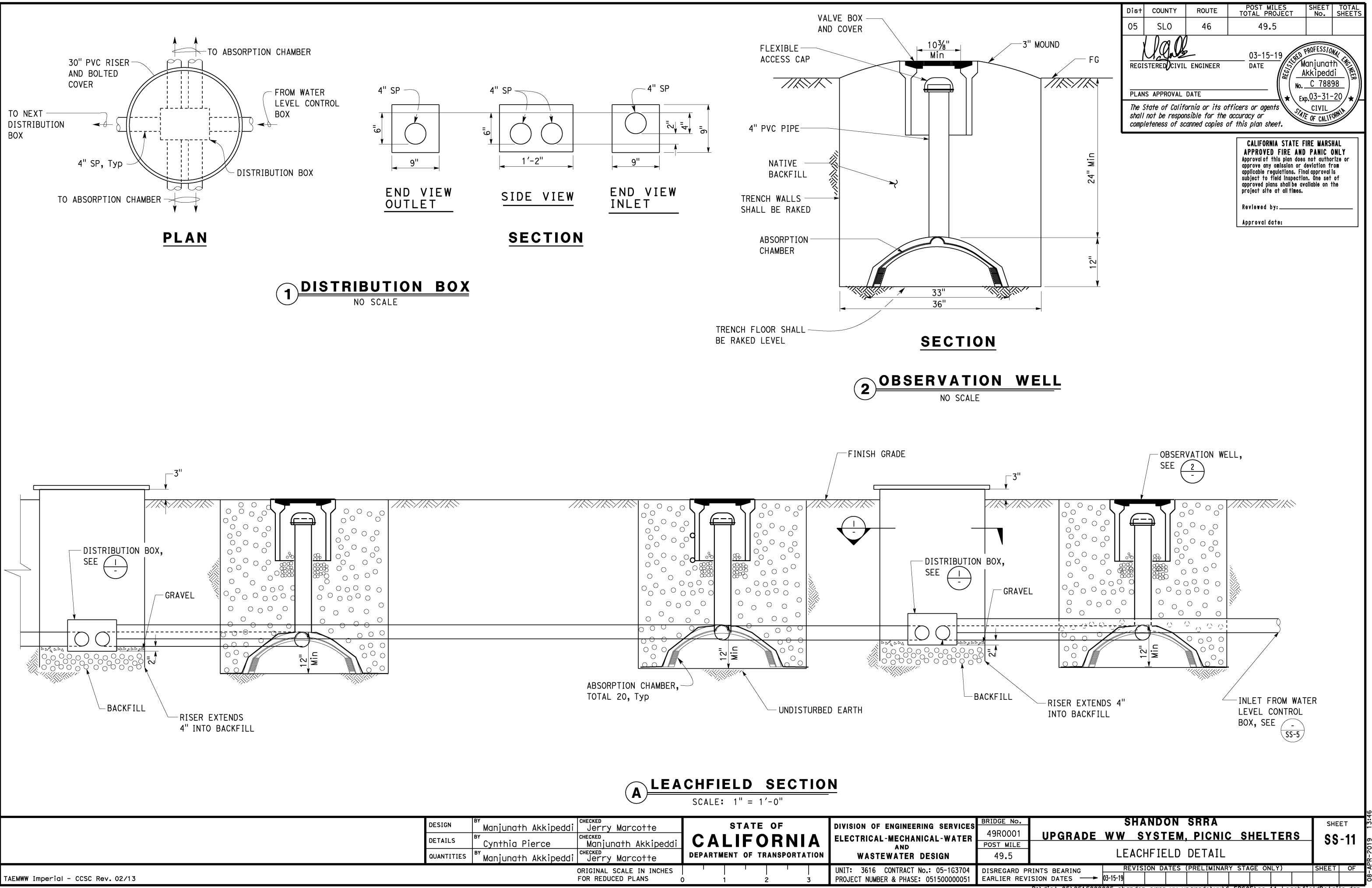
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LEACHFIELD PLAN
1" : 5'

- Note:
1. Observation wells shall be installed at the end of each absorption chamber.
 2. Distribution box outlet (to the next distribution box) I.E. and the outlets to the absorption chambers are 2" lower than the I.E., inlet to the distribution box.
 3. The absorption chambers for each distribution box are installed level at the same elevation.

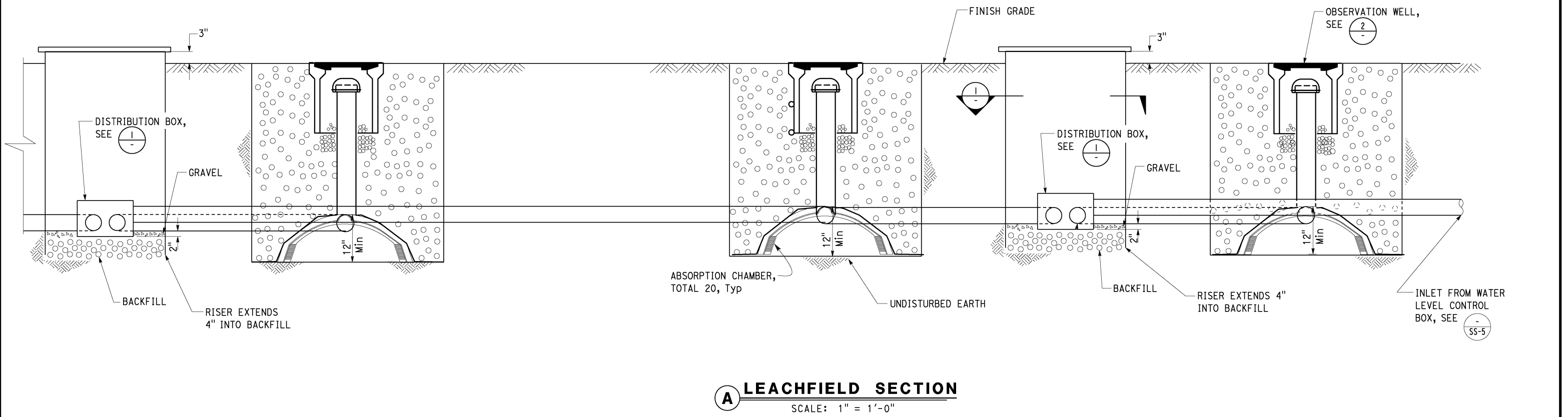


Dist	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		
REGISTERED CIVIL ENGINEER			03-15-19	DATE	
PLANS APPROVAL DATE					
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STATE OF CALIFORNIA



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	DETAILS	BY Cynthia Pierce	CHECKED Manjunath Akkipeddi			49R0001	UPGRADE WW SYSTEM, PICNIC SHELTERS								
	QUANTITIES	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte			POST MILE 49.5	LEACHFIELD DETAIL								
ORIGINAL SCALE IN INCHES FOR REDUCED PLANS			0 1 2 3			UNIT: 3616 CONTRACT No.: 05-1G3704 PROJECT NUMBER & PHASE: 051500000051		DISREGARD PRINTS BEARING EARLIER REVISION DATES → 03-15-19		REVISION DATES (PRELIMINARY STAGE ONLY)				SHEET	OF
TAEWW Imperial - CCSC Rev. 02/13															

Dist	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		


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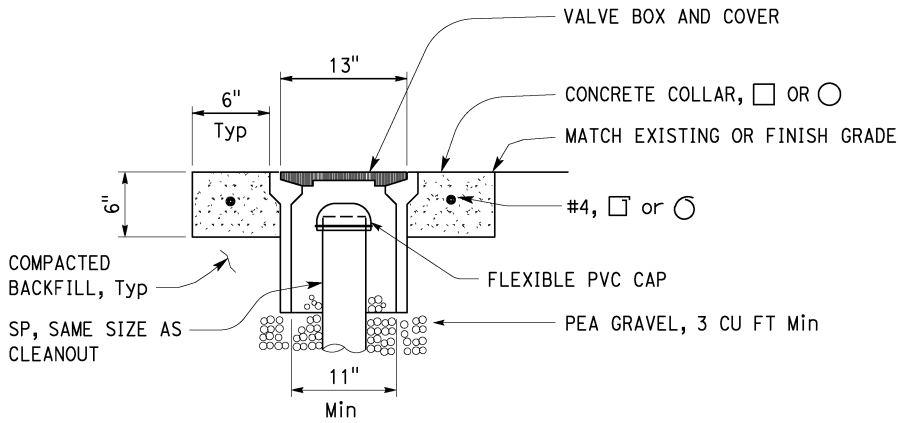
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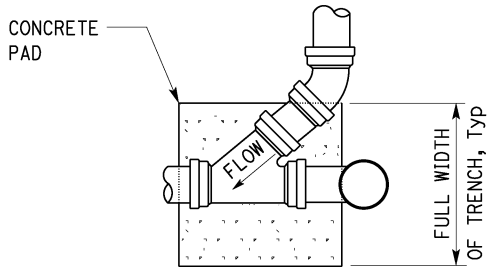
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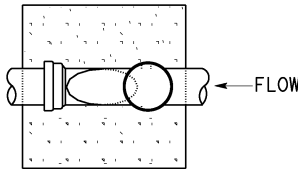
Reviewed by: _____
Approval date: _____



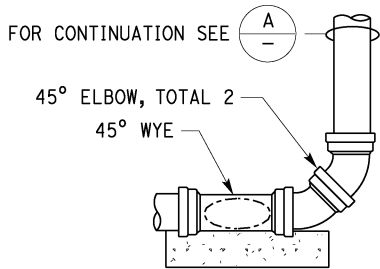
A **COTG IN VALVE BOX**
NO SCALE



PLAN

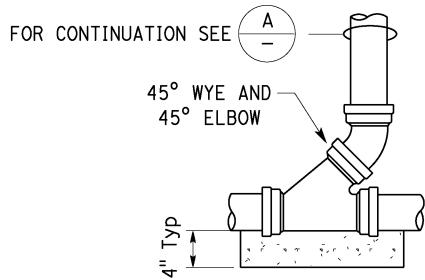


PLAN



ELEVATION

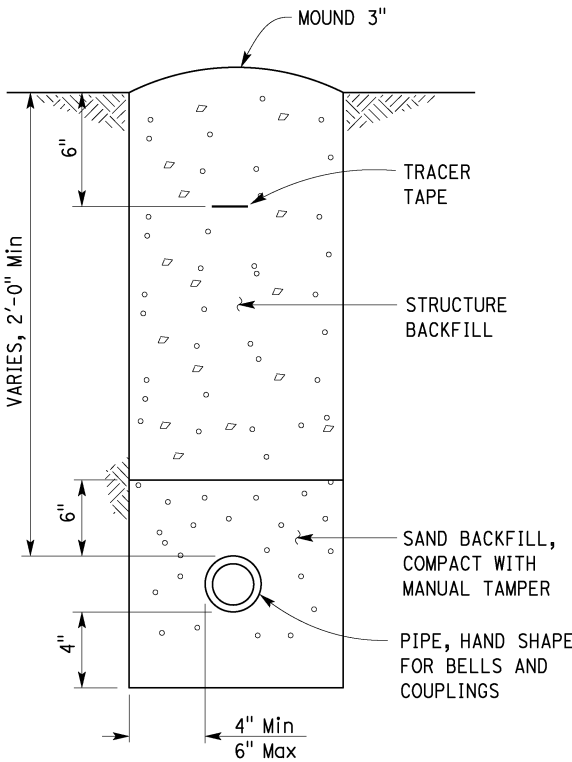
CLEANOUT (AT BEND)



ELEVATION

CLEANOUT (STANDARD)

1 **CLEANOUT TO GRADE**
NO SCALE



3 **SEWER / FORCE MAIN**
NO SCALE

DESIGN	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte
DETAILS	BY Cynthia Pierce	CHECKED Manjunath Akkipeddi
QUANTITIES	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte

STATE OF
CALIFORNIA
DEPARTMENT OF TRANSPORTATION

DIVISION OF ENGINEERING SERVICES
ELECTRICAL-MECHANICAL-WATER
AND
WASTEWATER DESIGN

BRIDGE No.
49R0001
POST MILE
49.5

SHANDON SRRA
UPGRADE WW SYSTEM, PICNIC SHELTERS
DETAILS

SHEET
SS-12

Dist	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
05	SLO	46	49.5		



03-15-19
DATE

REGISTERED CIVIL ENGINEER

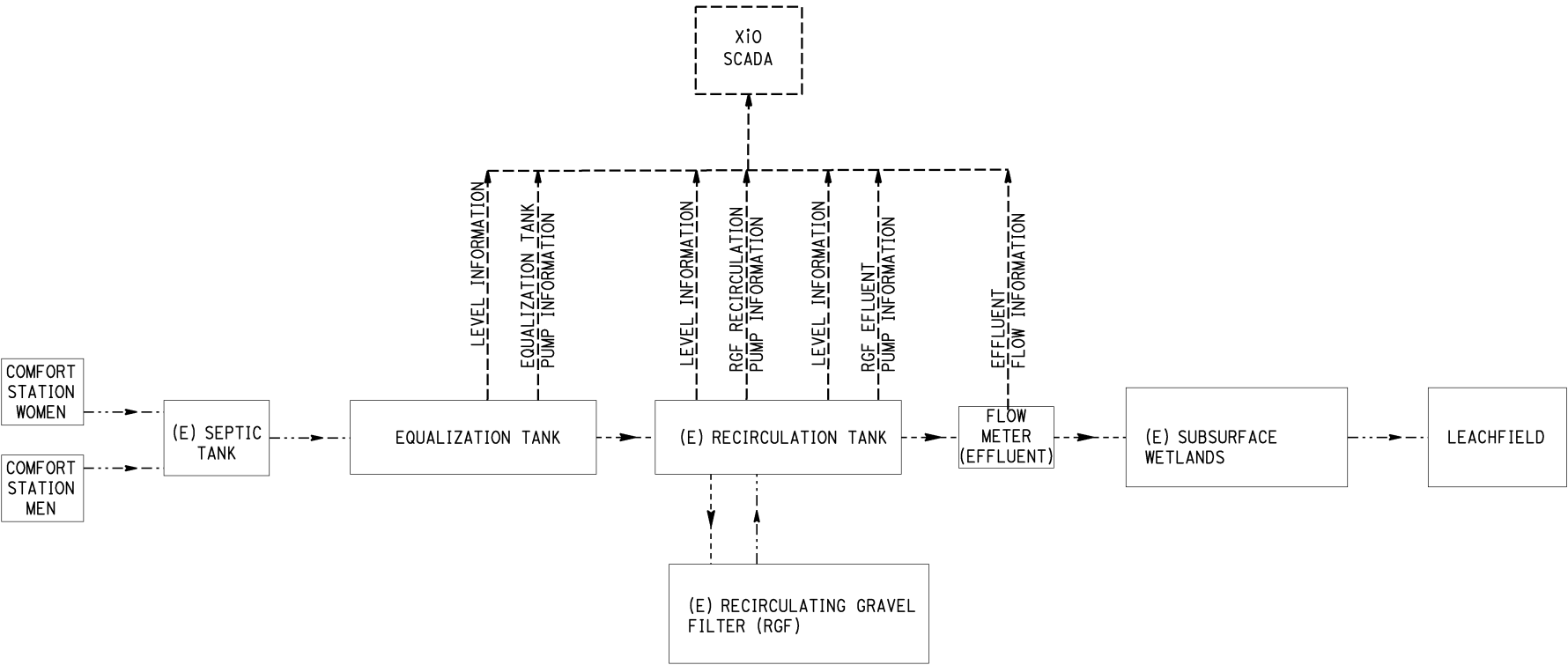
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Approval date: _____



WASTEWATER SYSTEM FLOW DIAGRAM

NO SCALE

- > FORCE MAIN
- ...> SEWER PIPE (GRAVITY)
- > LOW VOLTAGE CONDUCTOR/CONDUIT

THIS PLAN APPROVED FOR SANITARY SEWER WORK ONLY

	DESIGN	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte	STATE OF CALIFORNIA DEPARTMENT OF TRANSPORTATION	DIVISION OF ENGINEERING SERVICES ELECTRICAL-MECHANICAL-WATER AND WASTEWATER DESIGN	BRIDGE No.	SHANDON SRRA					SHEET SS-13				
	DETAILS	BY Cynthia Pierce	CHECKED Manjunath Akkipeddi			49R0001	UPGRADE WW SYSTEM, PICNIC SHELTERS									
	QUANTITIES	BY Manjunath Akkipeddi	CHECKED Jerry Marcotte			POST MILE 49.5	WASTEWATER SYSTEM FLOW DIAGRAM									
TAEWW Imperial - CCSC Rev. 02/13			ORIGINAL SCALE IN INCHES FOR REDUCED PLANS		0	1	2	3	UNIT: 3616 CONTRACT No.: 05-1G3704 PROJECT NUMBER & PHASE: 051500000051		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)		SHEET	OF
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